SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

ADVANCED TEST & AUTOMATION, INC.
1-32 Steeles Avenue East
Milton, ON L9T 5A1
Anthony Khoraych Phone: (647) 477-6247

CALIBRATION

Valid To: August 31, 2018 Certificate Number: 3976.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations[1]:

I. Electrical – DC/ Low Frequency

Parameter/Equipment	Range	CMC[2] (±)	Comments
DC Voltage – Generate[3]	(Up to 329.99) mV (0.333 to 3.29) V (3.30 to 32.99) V (30 to 329.99) V (100 to 1000) V	6 µV 31 µV 0.34 mV 4.8 mV 16 mV	Multi-function calibrator
DC Current – Generate[3]	(Up to 329.999) µA (0.33 to 3.29999) mA (3.3 to 32.9999) mA (33 to 329.999) mA (0.33 to 1.09999) A (1.1 to 2.99999) A (3.0 to 10.9999) A (11 to 20.5) A	54 nA 0.3 µA 2.8 µA 28 µA 0.21 mA 0.93 mA 4.7 mA 17 mA	Multi-function calibrator

Parameter/Equipment	Range	CMC[2] (±)	Comments
Resistance – Generate[3]	(Up to 10.9999) Ω (11 to 32.9999) Ω (33 to 109.9999) Ω (110 to 329.9999) Ω (330 to 1.099999) kΩ (1.1 to 3.299999) kΩ (3.3 to 10.99999) kΩ (11 to 32.99999) kΩ (33 to 109.9999) kΩ (110 to 329.99999) kΩ (330 to 1.099999) MΩ (1.1 to 3.299999) MΩ (3.3 to 10.99999) MΩ (11 to 32.99999) MΩ (33 to 109.9999) MΩ (110 to 329.9999) MΩ (330 to 1100) MΩ	1.1 mΩ 2.2 mΩ 4.6 mΩ 12 mΩ 37 mΩ 0.12 Ω 0.36 Ω 1.2 Ω 3.6 Ω 12 Ω 36 Ω 0.13 kΩ 0.4 kΩ 3.4 kΩ 8.4 kΩ 0.18 MΩ 0.93 MΩ	Multi-function calibrator
Capacitance – Generate[3]	(0.19 to 0.39) nF (0.4 to 1.09) nF (1.1 to 3.29) nF (3.3 to 10.99) nF (11 to 32.99) nF (33 to 109.99) nF (110 to 329.99) nF (0.33 to 1.09) µF (1.1 to 3.299) µF (3.3 to 10.99) µF (11 to 32.99) µF (33 to 109.99) µF (110 to 329.99) µF (0.33 to 1.09) mF (1.1 to 3.29) mF (3.3 to 10.99) mF (11 to 32.99) mF (33 to 110) mF	9.3 pF 12 pF 21 pF 30 pF 0.15 nF 0.3 nF 1.1 nF 3 nF 11 nF 30 nF 0.14 µF 0.47 µF 1.5 µF 4.7 µF 15 µF 47 µF 0.22 mF 1 mF	Multi-function calibrator

Parameter/Range	Frequency	CMC[2] (±)	Comments
AC Voltage – Generate[3]			Multi-function calibrator
(1 to 32.99) mV	(10 to 45) Hz (45 to 10) kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	25 µV 8.9 µV 10 µV 31 µV 99 µV 0.24 mV	
(33 to 329.99) mV	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	85 µV 44 µV 48 µV 97 µV 0.23 mV 0.57 mV	
(0.33 to 3.29) V	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.81 mV 0.43 mV 0.54 mV 0.81 mV 1.9 mV	
(3.3 to 32.99) V	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	8.2 mV 4.4 mV 6.7 mV 9.5 mV 24 mV	
(33 to 329.9) V	45 Hz to 10 kHz (1 to 10) kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	51 mV 52 mV 69 mV 86 mV 0.56 V	
(330 to 1020) V	45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.25 V 0.21 V 0.25 V	

Parameter/Range	Frequency	CMC[2] (±)	Comments
AC Current – Generate[3]			Multi-function calibrator
(29 to 329.99) µA	(10 to 20) Hz	0.82 µA	
	(20 to 45) Hz	0.46 µA	
	45 Hz to 1 kHz	0.4 µA	
	(1 to 5) kHz	0.89 µA	
	(5 to 10) kHz	2.2 µA	
	(10 to 30) kHz	4.4 µA	
(0.33 to 3.29) mA	(10 to 20) Hz	5.2 µA	
	(20 to 45) Hz	3.3 µA	
	45 Hz to 1 kHz	2.7 µA	
	(1 to 5) kHz	5.3 µA	
	(5 to 10) kHz	13 µA	
	(10 to 30) kHz	26 µA	
(3.3 to 32.99) mA	(10 to 20) Hz	48 µA	
	(20 to 45) Hz	25 µA	
	45 Hz to 1 kHz	12 µA	
	(1 to 5) kHz	22 µA	
	(5 to 10) kHz	54 µA	
	(10 to 30) kHz	0.11 mA	
(33 to 329.99) mA	(10 to 20) Hz	0.48 mA	
	(20 to 45) Hz	0.25 mA	
	45 Hz to 1 kHz	0.12 mA	
	(1 to 5) kHz	0.3 mA	
	(5 to 10) kHz	0.59 mA	
	(10 to 30) kHz	1.2 mA	
(0.33 to 1.09) A	(10 to 45) Hz	1.6 mA	
	45 Hz to 1 kHz	0.51 mA	
	(1 to 5) kHz	5.9 mA	
	(5 to 10) kHz	25 mA	
(1.1 to 2.99) A	(10 to 45) Hz	4.3 mA	
	45 Hz to 1 kHz	1.5 mA	
	(1 to 5) kHz	15 mA	
	(5 to 10) kHz	62 mA	
(3 to 10.99) A	(45 to 100) Hz	53 mA	
	100 Hz to 1 kHz	10 mA	
	(1 to 5) kHz	0.26 A	
(11 to 20.5) A	(45 to 100) Hz	23 mA	
	100 Hz to 1 kHz	28 mA	

Parameter/Equipment	Range	CMC[2] (±)	Comments
DC Voltage – Measure[3]	Up to 100 mV (0.1 to 1) V (1 to 10) V (10 to 100) V (100 to 1000) V	58 µV 17 µV 0.24 mV 58 mV 13 mV	Precision DMM
DC Current – Measure[3]	Up to 100 µA (0.1 to 1) mA (1 to 10) mA (10 to 100) mA (100 to 400) mA (0.4 to 1) A (1 to 3) A (3 to 10) A	28 nA 69 nA 58 µA 0.58 mA 78 µA 5.8 mA 5.8 mA 58 mA	Precision DMM
Resistance – Measure[3]	Up to 10 Ω (10 to 100) Ω (0.1 to 1) kΩ (1 to 10) kΩ (10 to 100) kΩ	3.5 mΩ 9.3 mΩ 56 mΩ 0.54 Ω 6 Ω	4 wire resistor with precision DMM
Resistance – Measure[3]	Up to 100 Ω (0.1 to 1) kΩ (1 to 10) kΩ (10 to 100) kΩ (0.1 to 1) MΩ (1 to 10) MΩ (10 to 100) MΩ (100 to 1000) MΩ	8.4 mΩ 56 mΩ 0.53 mΩ 5.9 Ω 0.11 kΩ 3.3 kΩ 0.13 MΩ 17 MΩ	2 wire resistor with precision DMM

Parameter/Range	Frequency	CMC[2] (±)	Comments
AC Voltage – Measure[3]			Precision DMM
Up to 100 mV	10 Hz 20 kHz 50 kHz 100 kHz 300 kHz	80 µV 24 µV 65 µV 0.44 mV 1.6 mV	
(0.1 to 1) V	10 Hz 20 kHz 50 kHz 100 kHz 300 kHz	0.77 mV 0.11 mV 0.41 mV 3.1 mV 4 mV	
(1 to 10) V	10 Hz 20 kHz 50 kHz 100 kHz 300 kHz	8.1 mV 2.1 mV 7.7 mV 21 mV 0.21 V	
(10 to 100) V	45 Hz 20 kHz 50 kHz 100 kHz	14 mV 13 mV 73 mV 0.18 V	
(100 to 1000) V	45 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz	0.48 V 0.48 V 0.14 V 0.13 V 0.44 V 1.3 V	
AC Current – Measure[3]			Precision DMM
Up to 100 µA	10 Hz 1 kHz 5 kHz 10 kHz	0.21 µA 0.19 µA 0.18 µA 0.29 µA	
(0.1 to 1) mA	10 Hz 1 kHz 5 kHz 10 kHz	1.9 µA 1.9 µA 1.3 µA 2.9 µA	
(1 to 10) mA	10 Hz, 1 kHz 5 kHz 10 kHz	23 µA 21 µA 12 µA 44 µA	

Parameter/Range	Frequency	CMC[2] (±)	Comments
AC Current – Measure[3] (cont)			Precision DMM
(10 to 100) mA	10 Hz 1 kHz 5 kHz 10 kHz	0.24 mA 0.21 mA 0.13 mA 0.37 mA	
(100 to 400) mA	10 Hz 1 kHz 5 kHz 10 kHz	1.6 mA 0.37 mA 0.46 mA 4.3 mA	
(0.4 to 1) A	45 Hz 1 kHz 5 kHz 10 kHz	0.85 mA 0.89 mA 1.1 mA 10 mA	
(1 to 3) A	45 Hz 1 kHz 10 kHz	1.9 mA 2 mA 19 mA	
(3 to 10) A	45 Hz 1 kHz	7.7 mA 11 mA	

Parameter/Equipment	Range	CMC[2] (±)	Comments
Capacitance – Measure[3]	(Up to 1) nF (1 to 10) nF (10 to 100) nF (0.1 to 1) μF (1 to 10) μF (10 to 100) μF (0.1 to 1) mF (1 to 10) mF (10 to 100) mF	14 pF 33 pF 0.39 nF 3.3 nF 33 nF 0.51 μF 7.6 μF 86 μF 1.2 mF	Precision DMM

II. Fluid Quantities

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Flow (Hydraulic)[3]	Up to 266 kg/min	0.07 %	Endress & Hauser Promass 83F DN25
	(266 to 866) kg/min	0.27 %	Sitrans FC Mass Flo Mass 2100

III. Mechanical

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Gauge Pressure[3] (Pneumatic & Hydraulic)	Up to 100 Bar	0.01 %	ASTM D5720
Torque[3]	(0 to 100) N.m	0.074 N.m	ASTM E2428

IV. Thermodynamics

Parameter/Equipment	Range	CMC[2] (±)	Comments
Temperature Sensors/ Resistance Temperature Detectors[3]	(-50 to 250) °C	0.06 °C	ASTM E644

V. Time & Frequency

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Rotational Speed[3]	Up to 10 000 RPM	0.09 %	ASTM E2658

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Frequency – Measuring Equipement[3]	(0.01 to 119.99) Hz (120 to 1199.9) Hz (1.2 to 11.999) kHz (12 to 119.99) kHz (120 to 1199.9) kHz (1.2 to 2) MHz	5.8 mHz 58 mHz 0.58 Hz 5.8 Hz 58 Hz 0.58 kHz	Multi-function calibrator
Frequency – Measure[3] (0.1 to 1) V	 Up to 10 Hz (10 to 40) Hz (0.04 to 300) kHz (300 to 1000) kHz	 7.3 mHz 7.4 mHz 59 Hz 75 Hz	Precision DMM

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[1] This laboratory offers commercial and field calibration service.

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

[3] Field calibration service is available for this calibration and this laboratory meets A2LA R104 – General Requirements: Accreditation of Field Testing and Field Calibration Laboratories for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

[4] In the statement of CMC, percentages are to be read as percent of reading, unless otherwise noted.

Accredited Laboratory

A2LA has accredited

ADVANCED TEST AND AUTOMATION, INC.

Milton, Ontario, CANADA

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).

Presented this 24th day of June 2016.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 3976.01
Valid to August 31, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.